# Fibre Coupling Project Report

Jadon Tsai, Geoffrey Cuff-Chartrand

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## Abstract

Fibre coupling is an important and tedious task in optical systems, where mirrors and optical instruments are aligned to maximize power transfer into a fibre optic. In this report, we derive a theoretical expression for fibre coupling efficiency by calculating the overlap integral between two Gaussian beams, using both positional and angular terms. Using this model, we explore both the adaptive coordinate descent and Nedler-Mead optimization algorithms for maximizing coupling efficiency in the presence of Gaussian noise. Numerical simulations are performed using realistic system parameters and added noise to emulate experimental conditions. In our model, we showed that adaptive coordinate descent reached a maxima of approximately 2.45 Watts within 20 iterations. By comparison, the Nelder-Mead simplex method reached a lower maximum power of approximately 2.22 W and required 46 iterations to do so. These results indicate that adaptive coordinate descent is both faster and more effective at locating the global maximum of the coupling function under the simulated noise conditions. Although experimental validation was not possible due to hardware limitations, our results demonstrate that adaptive optimization methods can reliably converge toward optimal coupling configurations and provide insight into the strengths and limitations of different algorithmic approaches for automated fibre coupling.

## 1 Introduction

Fibre coupling is the process of aligning mirrors and the fibre in a way that maximizes the amount of power transferred from a laser into a fibre. This tedious process can be greatly simplified by using a machine. [4]

Boris Braverman is a professor in the U of T department of physics. One of his students recently designed a device that can automatically adjust mirrors to allow for automated fibre coupling. In this project we design an algorithm that optimizes the fibre coupling. Unfortunately the equipment had technical issues beyond our control and significantly beyond the scope of this course, so we couldn't test the algorithms on the actual setup. The setup consists of the 750nm laser, which passes through a focusing lens, two 50/50 beam splitters, then in reflected by a mirror controlled by a machine into a waveguide leading to the fibre, which is positioned by another machine. The fibre is then connected to a photodiode which records the power transferred. The setup is shown in Figure 1.

The algorithm will focus on coupling once the laser is coupled to the point that changes in coupling are detectable after small changes in position. Prior to reaching this point, the

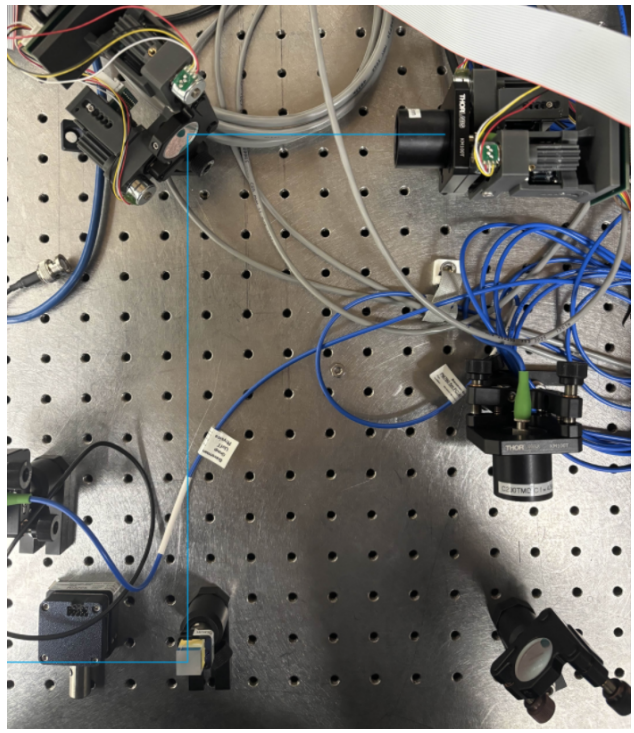

Figure 1: The laser setup, from the second beam splitter to the waveguide, where the blue line indicates the path the laser takes

algorithms will not function as they cannot detect any change in coupling when sampling. Reaching this point will have to be done by manual coupling with a guide laser.

We will first determine the theoretical relation between the angles and the coupling power. Then we will test various optimization functions on this function, adding in noise to approximate the noise of the setup.

## 2 Power function

To calculate the theoretical fibre coupling, we calculate the overlap integral between the beam that would be emitted by the fibre and the beam that arrives at the fibre from the mirrors. We assume both beams are Gaussian. We define the origin to be at the centre of the fibre outlet. Therefore, the beam which would be emitted by the fibre has electromagnetic field given by

$$e^{-(x^2+y^2)/w^2}$$

where $w$ is the width of the beam [1]. The path length of all considered paths are very close, so we neglect any change in beam width. The actual beam arrives offset from the centre of the fibre and at an angle. The centre of the real beam is at a position $(a, b)$. The angle formed between the real and ideal beam produces a phase shift depending on the position. Treating the one-dimensional case, we use figure 2. As seen in the figure, the distance between wavefronts is given by k. When the wavefront forms an angle $\theta$, the distance from the actual centre of the wavefront to location of intersection is $d \cos\theta$, while phase shift is

$k \sin\theta$. We can approximate $\cos\theta \approx 1$ while $\sin\theta \approx \theta$, as the angles we are dealing with are very small.

The integral is

$$A \int_{-\infty}^{\infty} e^{-(x^2+y^2)/w^2} * e^{-((x-a)^2+(y-b)^2)/w^2 - i(kx\theta_x + ky\theta_y)} dxdy$$

where $A$ is the amplitude of the beam, and the power is given by the square of this integral. We can analytically calculate this integral.

We can separate the integral into a integral only in $x$ and one only in $y$. Therefore, this integral is the multiple of two in the form

$$\int_{-\infty}^{\infty} e^{-(x^2/w^2)} * e^{-((x-a)^2/w^2 - i(kx\theta_x)} dx$$

Completing the square,

$$-2x^2/w^2 + 2xa/w^2 - a^2/w^2 - i(kx\theta_x)$$

$$= -2/w^2(x - \frac{2a/w^2 + ik\theta_x}{4/w^2})^2 - a^2/w^2 + \frac{(2a/w^2 - ik\theta_x)^2}{8/w^2}$$

Therefore, the integral is

$$C * \int_{-\infty}^{\infty} e^{-2/w^2(x - \frac{2a/w^2 - ik\theta_x}{4/w^2})^2} dxdy$$

where

$$C = e^{-a^2/w^2 + \frac{w^2(2a/w^2 - ik\theta_x)^2}{8}}$$

The integral part is a standard Gaussian integral and is equal to $\sqrt{\frac{w^2\pi}{2}}$. Therefore, the overall power transfer is

$$P = (\frac{A\pi w^2}{2} e^{-a^2/w^2 + \frac{w^2(2a/w^2 - ik\theta_x)^2}{8} - b^2/w^2 + \frac{w^2(2b/w^2 - ik\theta_y)^2}{8}})^2$$

Now we connect $a, b, \theta_x, \theta_y$ to the angles formed by the mirror and coupling mount. We consider just the x-axis, with the calculations for the y axis being similar. The geometry of the setup is given by figure 2. As shown, the mirror is positioned at $(0, 0)$ and the angle formed between the mirror and the x is $\theta_{1x}$. The fibre is positioned at $(c, 0)$, and forms the angle $\theta_{2x}$ with the x-axis. The initial angle of the beam is given by $\theta_{ix}$. The angle of between the initial and reflected beams is equal to twice the angle of incidence, and is therefore $180 - 2\theta_{1x} + 2\theta_{ix}$. Therefore, the smallest angle of the reflected beam with the x-axis is $180 - 180 + 2\theta_{1x} - 2\theta_{ix} + \theta_{ix} = 2\theta_{1x} - \theta_{ix}$. Depending on the angle of the mirror, the position at the laser reaches the mirror will change. However, this effect is much less significant then the change in angle and so will be neglected. Assuming $\theta_{2x}$ is close to 90°, and $2\theta_{1x} - \theta_{ix}$ is close to zero, which they must be for coupling to occur, the beam reaches the waveguide at $c, c(2\theta_{1x} - \theta_{ix})$. Therefore, the positional offset in the x direction is given

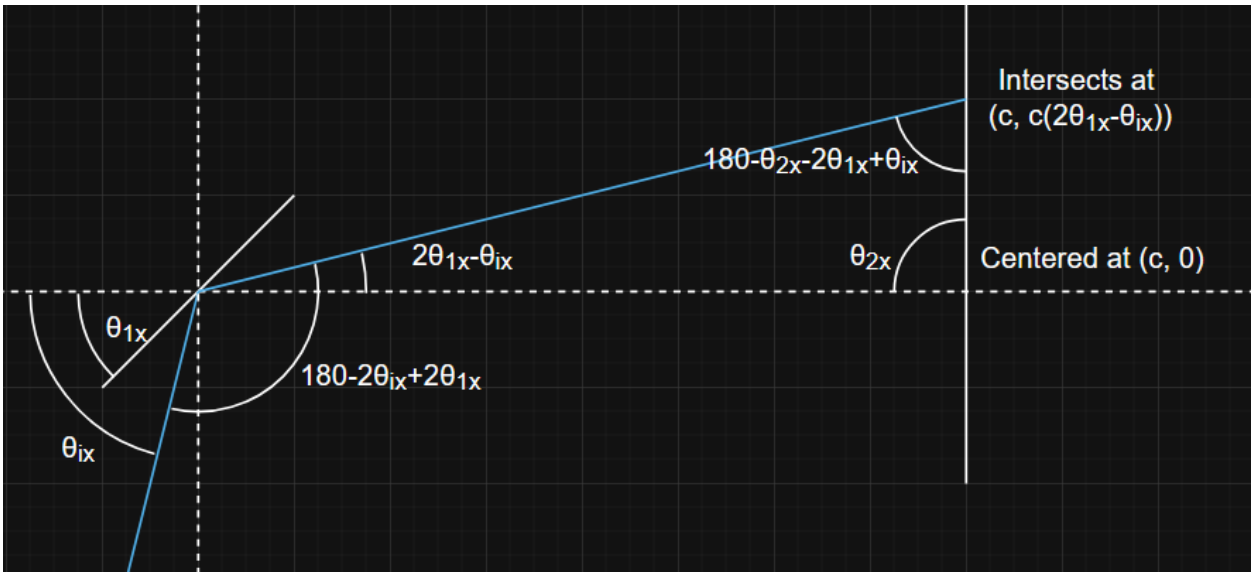


Figure 2: A diagram showing the relevant angles when connecting the angles of the mirrors to the positional and angular offset

by $c(2\theta_{1x} - \theta_{ix})$ while the angle offset is $180° - 2\theta_{1x} - \theta_{2x} + \theta_{ix}$, using the sum of angles of a triangle. Therefore the angle offset from the normal is $2\theta_{1x} + \theta_{2x} - \theta_{ix} - 90°$

The same relations apply to the y-axis. Combining these expressions, we obtain

$$P = (\frac{A\pi w^2}{2} e^{-a^2/w^2 + \frac{w^2(2a/w^2 - ik\theta_x)^2}{8} - b^2/w^2 + \frac{w^2(2b/w^2 - ik\theta_y)^2}{8}})^2 \tag{1}$$

$$a = c(2\theta_{1x} - \theta_{ix}) \tag{2}$$
$$b = c(2\theta_{1y} - \theta_{iy}) \tag{3}$$

$$\theta_x = 2\theta_{1x} + \theta_{2x} - \theta_{ix} - 90° \tag{4}$$
$$\theta_y = 2\theta_{1y} + \theta_{2y} - \theta_{iy} - 90° \tag{5}$$

## 2.1 Gradient ascent

Gradient ascent is the method of calculating the gradient, then moving some step in the direction of the gradient. There are theoretically no local maximums in our system, which motivated the use of gradient ascent. The noise-free gradient ascent iteration is

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k + \eta_k \nabla P(\boldsymbol{\theta}_k), \tag{6}$$

where $\eta_k$ is the step size, and P is the power function of angle $\theta$ [7]. However, we do not have an analytical solution for $\nabla P(\boldsymbol{\theta}_k)$, and we also need to factor in the noise of the real optical system. This creates presents the problem with standard gradient ascent; particularly with our narrow power function. Large step sizes would be required to come close to the peak, however they amplify noise at a rate proportional to the square of the step size [5]. Smaller step sizes would be impractical to use for our purposes, as they would take far too long.

### 2.1.1 Adaptive gradient ascent

To improve upon gradient descent, we used adaptive gradient descent (also known as adaptive cooridnate descent), which has the same conceptual background, but we adjust the step size depending on our proximity to a maxima [3]. We adjust the step size inversely to the observed gradient at a point; a parameter with large gradients at a point would have a smaller step size. This is similar to performing variance normalization [5]; the adaptive step size reduces the influence of high variance/noise points.

## 2.2 Nedler-Mead

Near the peak, the gradient values are lower, causing gradient ascent to become unreliable. By contrast, Nedler-Mead has the advantage of not taking any derivatives. It works by creating and maintaining a simplex, which is the simplest possible shape with n+1 vertices in n dimensions. [6] In our case, we have a 4 dimensional simplex with 5 vertices. Each vertex is a point in the parameter space with a value of the power function evaluated at that point. After each evaluation of the values at the vertices, we have 4 options; reflection, expansion, contraction, or shrinkage. We first test reflection by measuring the centroid of the simplex [2]. The vertex with the lowest power value is then reflected across the centroid of the simplex to create a new vertex. Let $x_w$ be the worst point, $x_s$ be the second worst point, $x_r$ be the reflected point, $x_c$ be the centroid and $x_b$ be the best point. With the heuristic that we want the highest possible x value, there are 4 possibilities and 4 different steps to take:

1. If $f(x_s) < f(x_r) < f(x_b)$, $x_r$ replaces $x_w$ and the entire simplex is essentially reflected.

2. if $f(x_r) > f(x_b)$, the simplex is "expanded" by moving further in the direction of reflection (i.e. we evaluate a point collinear with $x_r$ and $x_w$ that is further away from the simplex than $x_r$), as the function is showing significant improvements in that direction.

3. If $f(x_w) < f(x_r) < f(x_s)$, we know that the direction of reflection is promising, but the step size is too large. We instead perform "outside contraction", where we pick a point collinear and in between $x_r$ and $x_c$ as the new candidate point.

4. If $f(x_r) < f(x_w)$, then we are probably very close to a maxima, since the value close to the higher value points is worse than the worst point. We thus pick a candidate point that is collinear and in between $x_w$ and $x_c$ as the new candidate point.

These steps select candidate points which are then evaluated. If these candidate points do not provide improvements in power, the simplex is shrunk by moving all points towards the best point, which stays fixed. This happens because the current simplex geometry does not work, so we focus locally on the best candidate point.

# 3 Method

To test the various methods, we use the power function (1). We start the various methods from the same position, and graph the recorded amplitude as a function of the number of iterations. We use the value $c = 0.1m, A = 1, w = 0.001m, \theta_i = 90°$, which approximates our setup.

# 4 Results and discussion

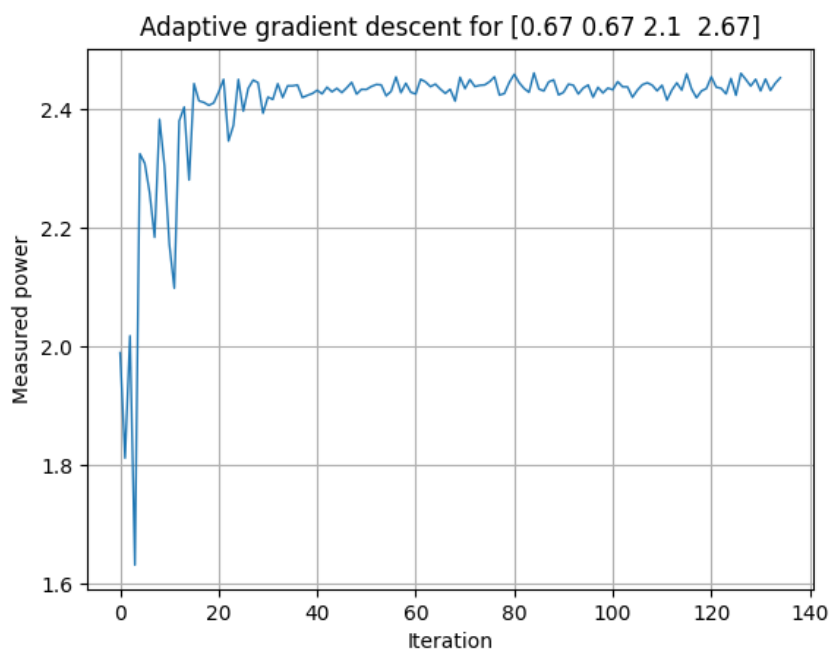


Figure 3: The results of power after 300 iterations of the adaptive coordinate descent method. The starting point was chosen to be at a point that has a non-negligible power reading

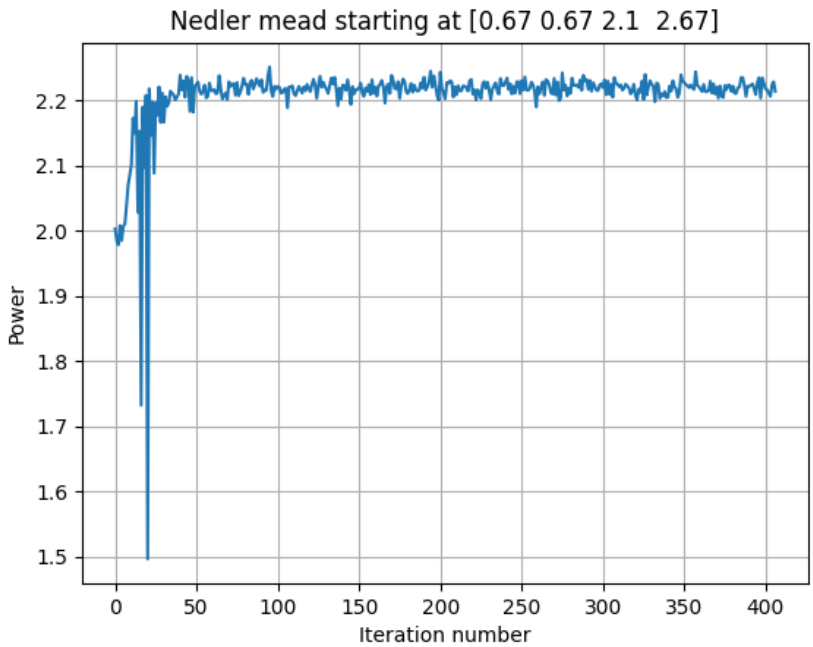


Figure 4: The results of power after 300 iterations of the Nedler-Mead method. The starting point was chosen as the same starting point for gradient descent to compare directly.

We found that gradient ascent was the more effective algorithm. It reached a peak of 2.45W as opposed to the Nedler-Mead which read a peak of 2.22V. Additionally, it reached its peak in less time, in only 20 iterations as opposed to Nedler-Mead's 46 iterations. This indicates that gradient ascent should be used as the fibre coupling algorithm.

Some differences with the actual system could result is issues when applying our algorithms. Imperfections in any of the optical components could result in a non-Gaussian beam, leading to local maximums not predicted by the theoretical power formula. This could result in our algorithms getting trapped in the local maximums, and not reaching the global maximum. Additionally, there is some noise in positioning, as the robot cannot perfectly position itself, that we did not consider.

Additionally, an algorithm to speed up finding a point where the power transfer is non-negligible would be useful. A version of this would be an algorithm that simply scans one axis at a time until it finds a non-negligible power reading, at which point another algorithm activates. This algorithm would need to be aware of local maximums, and so would need to be tuned based on the setup to avoid them.

To further improve the algorithms, we could attempt to include properties of the system. For example, moving the pairs of x-axis angles together could allow us to manipulate only the $a$ or $\theta_x$ terms, allowing for algorithms to optimize with those terms. This could be benifical, because at high $k$ values, the $\theta_x$ term is much more sensitive then then the $a$ term, and so different hyperparameters could be used.

# 5 Conclusion

In conclusion, using gradient ascent is functional for fibre-coupling, provided there are non-negligible readings. It takes around 20 iterations to reach the maximum. It is more effective then Nedler-Mead optimization. In the future, testing the algorithms on the proper setup and testing algorithms that take advantage of the properties of the system would be useful. Despite these limitations, the results demonstrate that adaptive optimization algorithms can significantly improve the efficiency and reliability of fibre coupling compared to manual alignment.

# References


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